HARDWARE & FIRMWARE DOCUMENTATION

IsdaSafe Pond Monitoring & Cooling System

ESP32-based water quality monitoring controller

Implementation Guide & Wiring Diagrams

Prepared for IsdaSafe Hardware Codes • Sketches: sketch1_aug1, DO_Calibration • 2026-08-01

1. System Overview

The IsdaSafe controller is a single ESP32 development board that continuously measures pond water quality and runs a temperature-triggered cooling fan. It combines an I2C sensor/display bus, an RS485 Modbus sensor bus, a single-wire humidity sensor, and a motor driver on one board, all driven by one firmware sketch.

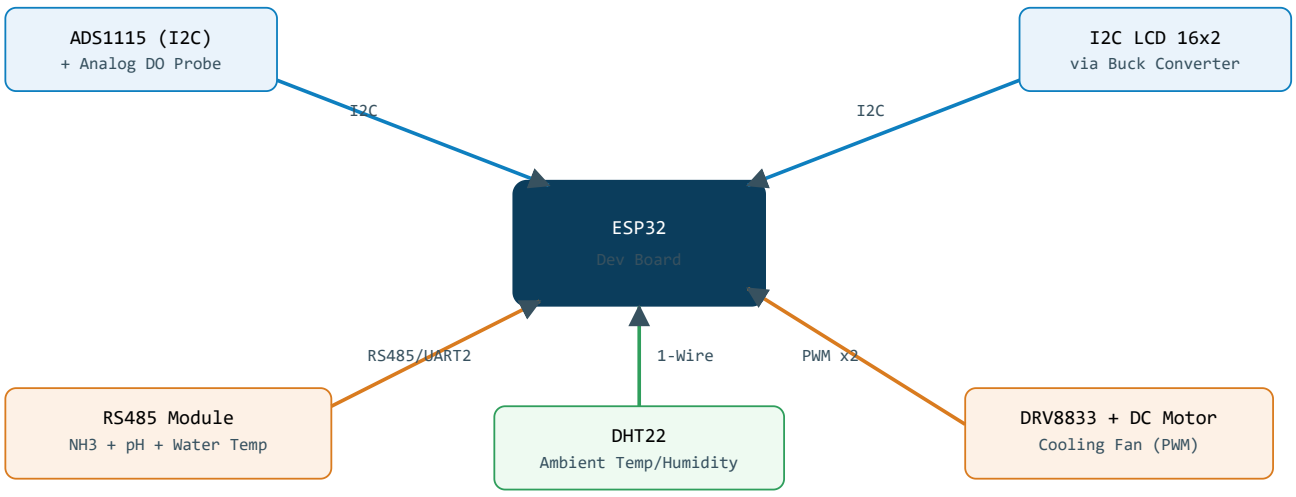
1.1 Measured Parameters

Parameter	Sensor	Interface	Notes
Dissolved Oxygen	Gravity Analog DO Sensor (SEN0237-A style)	Analog → ADS1115 (I2C)	Requires probe calibration, see Section 6
pH	RS485 Water Quality Ammonia & pH Sensor	RS485 / Modbus RTU	Same physical probe as Ammonia & Water Temp
Ammonia (NH3-N)	RS485 Water Quality Ammonia & pH Sensor	RS485 / Modbus RTU	Same probe, register 1
Water Temperature	RS485 Water Quality Ammonia & pH Sensor	RS485 / Modbus RTU	Same probe, register 2; used for DO compensation & fan control
Ambient Humidity / Air Temp	DHT22	Single-wire digital	Enclosure/ambient condition, not water

1.2 Outputs

Output	Module	Behavior
16x2 LCD Display	I2C LCD (PCF8574 backpack), powered via buck converter	Rotates 3 pages every 3s: DO/pH → NH3/Water Temp → Air Temp/Humidity
Cooling Fan	DRV8833 driving 3V / 1000RPM DC motor	PWM ramps 0→100% duty as water temp rises from 30°C to 34°C
Serial Log	USB serial, 115200 baud	Prints all readings once per poll cycle (3s)

1.3 Block Diagram



2. ESP32 Pin Assignment

ESP32 GPIO	Function	Connects To
GPIO21	I2C SDA	ADS1115 SDA, LCD SDA (shared bus)
GPIO22	I2C SCL	ADS1115 SCL, LCD SCL (shared bus)
GPIO15	Digital in (1-wire)	DHT22 data pin
GPIO16 (RX2)	UART2 RX	RS485 module Blue/RX wire
GPIO17 (TX2)	UART2 TX	RS485 module Green/TX wire
GPIO25	PWM out	DRV8833 AIN1
GPIO26	PWM out	DRV8833 AIN2
GPIO27	Digital out	DRV8833 nSLEEP (STBY), driven HIGH to enable

2.1 I2C Device Addresses

Device	Address	Notes
ADS1115	0x48	ADDR pin tied to GND (default)
I2C LCD backpack	0x27	Try 0x3F if the display stays blank on your backpack batch

2.2 RS485 Bus

Setting	Value
Protocol	Modbus RTU, function 0x03 (Read Holding Registers)
Baud rate	4800 (fixed by sensor)
Frame	8 data bits, no parity, 1 stop bit (8N1)
Slave address	1 (sensor's fixed Modbus address)
Registers read	0x0000-0x0002 → NH3, pH, Water Temp (one transaction)

> **Note:** The Gravity Active Isolated RS485-to-UART module auto-switches transmit/receive direction, so no DE/RE GPIO is wired or toggled in firmware.

3. Component Photo Gallery

Actual product photos for every module referenced in this build, for quick visual identification against what you have on the bench.



ESP32 Dev Board
FireBeetle ESP32 (DFR0478)



Analog DO Sensor
SEN0237-A + signal board



ADS1115 ADC Module
DFR0553



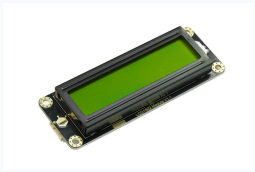
RS485 NH3/pH/Temp Probe
Water Quality Sensor



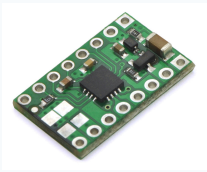
RS485-to-UART Adapter
Active Isolated, DFR0845



DHT22 Sensor
SEN0137



16x2 I2C LCD
Gravity LCD1602 (DFR0556)



DRV8833 Motor Driver
Dual H-bridge carrier



3V DC Motor
Small hobby / gearbox motor

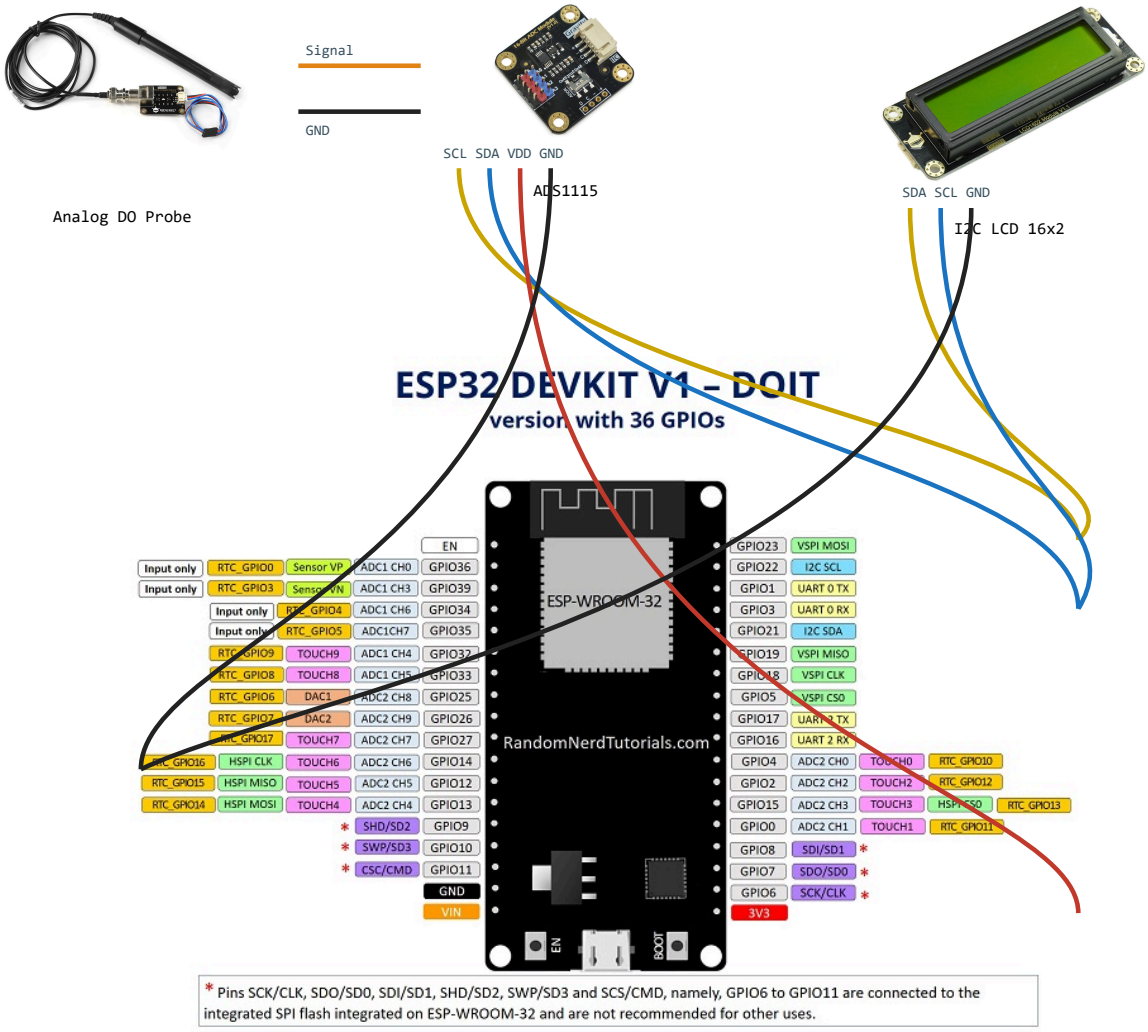


Buck Converter
LM2596-style step-down (LCD supply)

Photos sourced from official manufacturer/distributor product pages (dfrobot.com, pololu.com, adafruit.com, addicore.com) for component identification purposes only; exact motor/buck-converter appearance may vary by supplier.

4. Wiring Diagram — I2C Bus (DO Sensor + LCD)

The ADS1115 (feeding the analog DO probe) and the I2C LCD share the same two-wire I2C bus off the ESP32. Both devices are 3.3V-logic tolerant on this bus; the LCD's own backlight/module power comes from the buck converter, not from the ESP32.



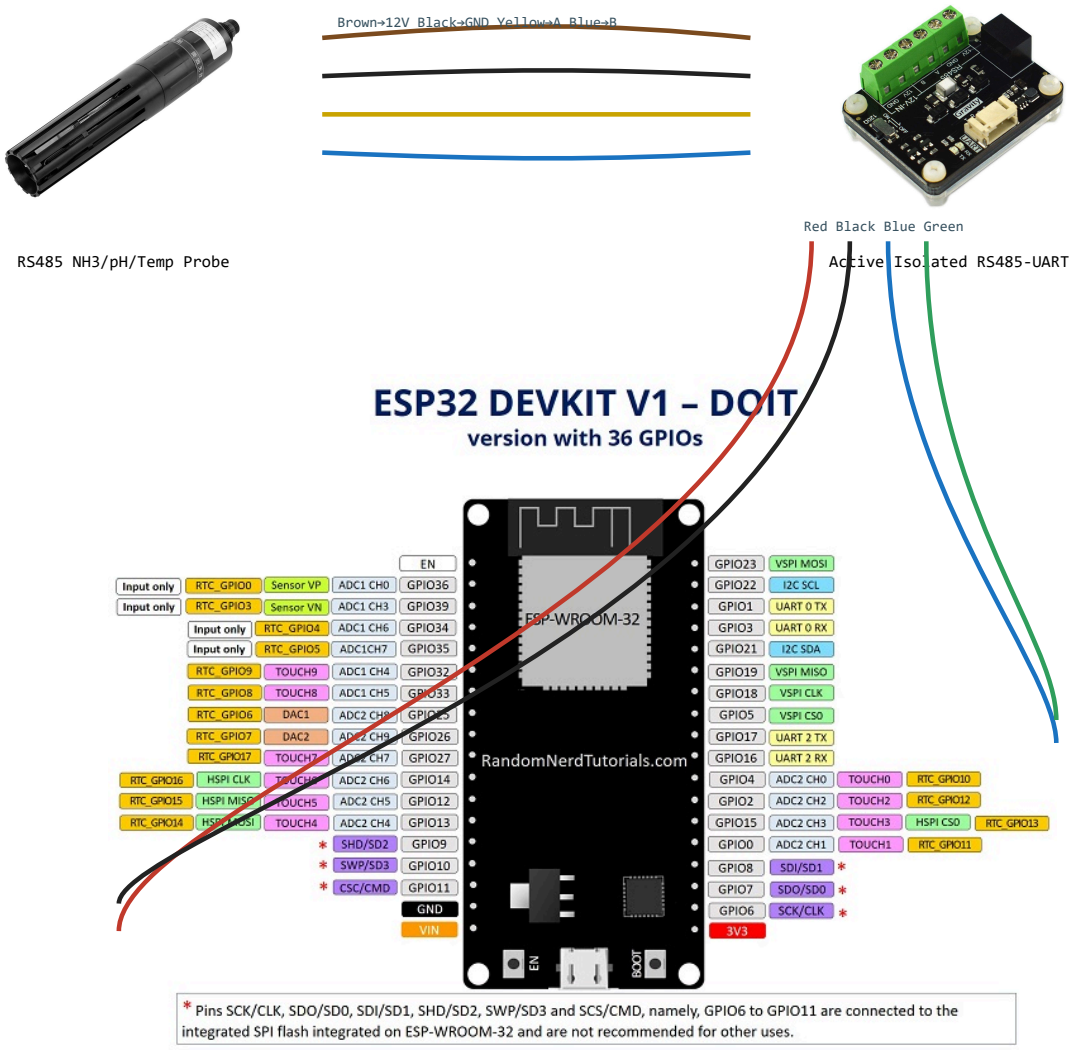
3.3V / Power — GND — SDA — SCL — DO Analog Signal

> **LCD power:** the LCD's VCC comes from the buck converter output (stepped down for the display), not directly from the ESP32 rail — only SDA/SCL/GND are shared with the ESP32 bus.

> **If the LCD stays blank:** run an I2C scanner sketch first — some PCF8574 backpacks ship at address 0x3F instead of 0x27.

5. Wiring Diagram — RS485 Water Quality Sensor

The combined Ammonia/pH/Water-Temp probe connects to the Gravity Active Isolated RS485-to-UART module, which in turn connects to the ESP32's UART2. The probe needs its own 10–30V DC supply (independent of the ESP32/module logic supply).



Brown = Probe 12V Black = GND Yellow = RS485 A Blue = RS485 B / module RX→GPIO16

Green = module TX→GPIO17 Red = 5V/VIN

> **Note:** the sensor probe's Brown wire needs its own 10–30V DC supply — it is not powered from the ESP32/adapter's 5V rail.

5.1 Protocol Summary

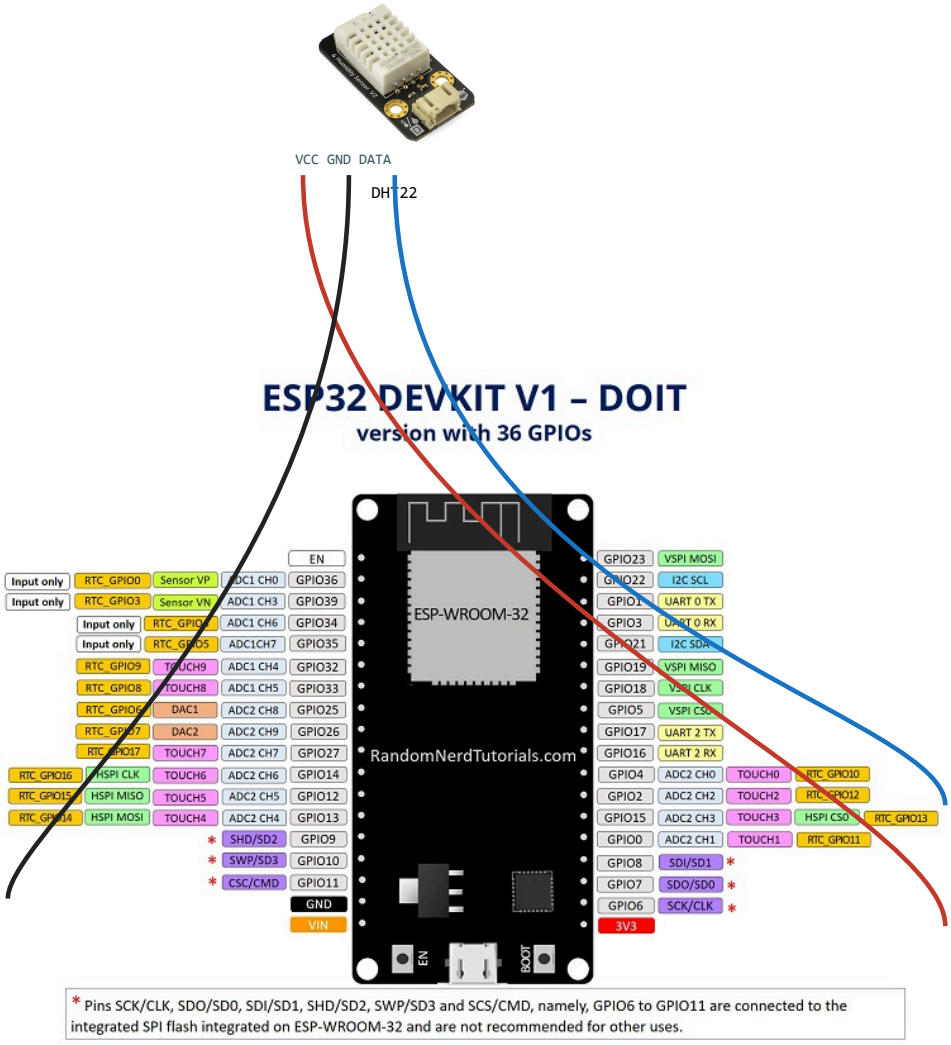
The main sketch polls this probe as a single Modbus transaction (function 0x03, start register 0x0000, quantity 3) equivalent to the request frame documented for the sensor:

```
01 03 00 00 00 03 05 CB   // slave 1, read 3 holding registers from 0x0000
```

Response layout: `[addr][func][bytecount][NH3 hi/lo][pH hi/lo][Temp hi/lo][CRC hi/lo]`, scaled as NH3÷100 (mg/L), pH÷100, Temp÷10 (°C). The `ModbusMaster` library builds and CRC-verifies this frame; the RS485 wiring/protocol matches DFRobot's official "Ammonia Nitrogen, pH & Temperature Monitoring" application note.

6. Wiring Diagram — DHT22 & Cooling Fan (DRV8833)

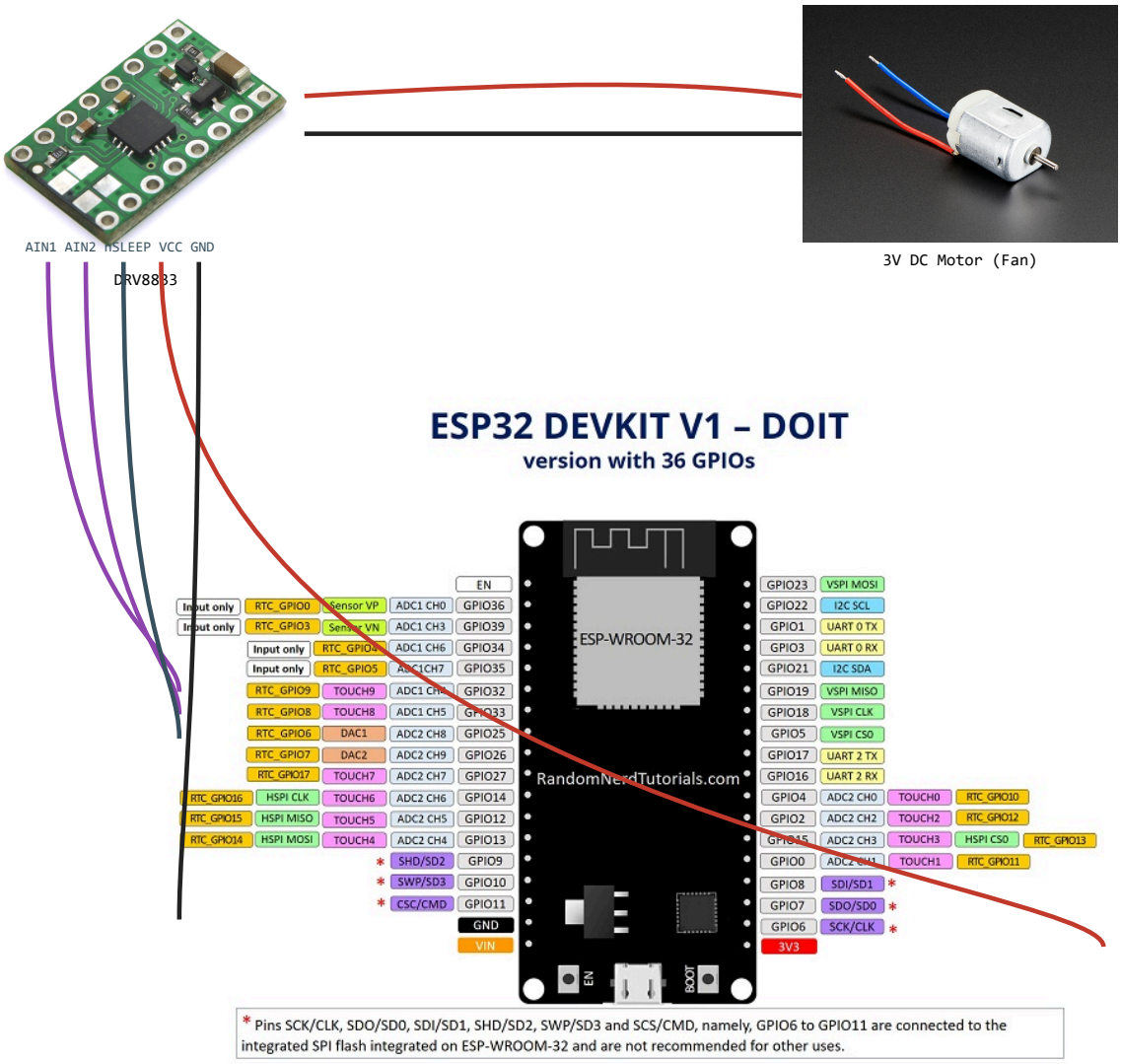
6.1 DHT22



VCC (3V3) GND DATA → GPIO15

DATA (GPIO15) needs a 4.7k–10kΩ pull-up resistor to 3.3V if the DHT22 breakout doesn't already include one.

6.2 DRV8833 + Cooling Motor



AIN1/AIN2 PWM → GPIO25/26 · nSLEEP → GPIO27 · VCC (3V3) / Motor + · GND / Motor -

> **Power note:** feed the DRV8833's `VM` pin from a supply matched to the 3V motor (not the ESP32's 3V3 logic rail directly, which can't source enough current) — a small dedicated 3-5V regulator or the buck converter's output is appropriate. `VCC` (logic) stays on 3.3V from the ESP32.

7. Dissolved Oxygen Probe Calibration



Gravity Analog DO Sensor

Run this once before first use, and again whenever the probe is replaced or has been in service for a long time. Use the standalone `DO_Calibration` sketch (ESP32 + ADS1115 equivalent of DFRobot's SEN0237-A calibration example) — it is a temporary sketch, flashed only for this step.

7.1 Procedure

Step	Action
1	Flash `DO_Calibration.ino` to the ESP32 and open Serial Monitor at 115200 baud.
2	Wet the probe in pure water, shake off excess drops, then expose it to air with normal air flow (no fan).
3	Wait for the printed voltage to stabilize; record that `Voltage(mv)` value and the current water/air temperature.
4 (two-point only)	Repeat steps 2-3 at a second, clearly different temperature (e.g. a chilled vs. a warmed cup of saturated water, both below 40°C).
5	Enter the recorded values into `sketch1_aug1.ino` as `DO_CAL1_V` / `DO_CAL1_T` (and `DO_CAL2_V` / `DO_CAL2_T` for two-point).
6	Set `DO_TWO_POINT_CALIBRATION` to `1` for two-point mode, or `0` to use single-point with a fixed 35mV/°C slope.
7	Re-flash `sketch1_aug1.ino` (the permanent operating sketch) to the ESP32.

> **Why two-point:** since the cooling fan actively changes water temperature, a single fixed-slope calibration drifts as temperature swings. Two-point calibration interpolates the saturation voltage between two known temperatures for better accuracy across that range.

7.2 Calibration Constants Reference

```
#define DO_TWO_POINT_CALIBRATION 1

#define DO_CAL1_V   1600   // mV reading at calibration point 1
#define DO_CAL1_T   25     // water temp (C) at calibration point 1
#define DO_CAL2_V   1300   // mV reading at calibration point 2
#define DO_CAL2_T   15     // water temp (C) at calibration point 2
```

8. Software Implementation

8.1 Sketches

Sketch	Purpose	Lifecycle
`sketch1_aug1/sketch1_aug1.ino`	Main operating firmware: polls all sensors, drives the LCD and cooling fan	Permanent — stays on the board
`DO_Calibration/DO_Calibration.ino`	Prints raw ADS1115 counts + mV for DO probe calibration	Temporary — flash only during calibration, then re-flash the main sketch

An ESP32 runs exactly one sketch at a time, so calibration and normal operation are two separate upload steps rather than two programs running together.

8.2 Required Libraries

Library	Used for
Adafruit ADS1X15	Reading the DO probe through the ADS1115
LiquidCrystal I2C	Driving the 16x2 I2C LCD
DHT sensor library (+ Adafruit Unified Sensor)	Reading the DHT22
ModbusMaster	RS485 Modbus RTU transactions with the water quality probe

8.3 Main Loop Behavior

Task	Interval	Notes
Sensor poll (RS485 + DO + DHT22)	3000 ms	Also logs a summary line to Serial and re-evaluates cooling duty
LCD page rotation	3000 ms	Cycles: DO/pH → NH3/Water Temp → Air Temp/Humidity
Cooling fan PWM	Recomputed every poll	0% below 30°C, ramps linearly to 100% at 34°C, floor duty ensures reliable spin-up

8.4 Key Functions (sketch1_aug1.ino)

Function	Role
`readWaterQualitySensor()`	Single Modbus read of NH3, pH, Water Temp from the RS485 probe
`readDissolvedOxygen(waterTempC)`	Reads ADS1115 voltage and applies the (single- or two-point) calibrated, temperature-compensated DO formula
`pollSensors()`	Orchestrates all sensor reads once per cycle
`controlCooling(waterTempC)`	Maps water temperature to DRV8833 PWM duty cycle
`updateLcd()`	Renders the current rotating LCD page

9. Bill of Materials

Photo	Qty	Item
	1	ESP32 development board
	1	Gravity: Analog Dissolved Oxygen Sensor (SEN0237-A style probe)
	1	ADS1115 16-bit I2C ADC module
	1	RS485 Water Quality Ammonia Nitrogen & pH Sensor (integrated NH3/pH/Temp probe)
	1	Gravity: Active Isolated RS485-to-UART Signal Adapter Module
	1	DHT22 temperature/humidity sensor
	1	16x2 I2C LCD (PCF8574 backpack)
	1	Buck converter (for LCD power)
	1	DRV8833 dual H-bridge motor driver module
	1	3V, 1000RPM DC motor (cooling fan)
—	1	10-30V DC power supply (RS485 probe)
—	1	4.7k–10kΩ resistor (DHT22 pull-up, if not on breakout)